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United States Patent Application Publication

20250279612

Kind Code

A1

Publication Date

September 04, 2025

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LEVER-TYPE CONNECTOR

Abstract

In a lever-type connector **10**, a lever **30** includes an arm portion **32** to be disposed along a side surface of a connector housing **20** and a connection detecting member **51** mounted on the arm portion **32** and displaceable between a standby position and a detection position. The arm portion **32** and the connection detecting member **51** include movement restricting portions **37**, **67** for restricting the connection detecting member **51** in an inclined posture from being displaced toward the standby position by locking each other if the connection detecting member **51** is set in the inclined posture by applying a force to the arm portion **32** in a direction to release connection and for allowing a displacement of the connection detecting member **51** if the connection detecting member **51** is set in the proper posture.

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Family ID: 88518511

Appl. No.: 18/859169

Filed (or PCT Filed): April 13, 2023

PCT No.: PCT/JP2023/014997

Foreign Application Priority Data

JP 2022-073117

Apr. 27, 2022

Publication Classification

Int. Cl.: H01R13/629 (20060101)

U.S. Cl.:

Background/Summary

TECHNICAL FIELD

[0001] The present disclosure relates to a lever-type connector.

BACKGROUND

[0002] A lever-type connector is provided with a lever for assisting connection to and separation from a mating connector by the principle of leverage. In connecting the lever-type connector to the mating connector, the both connectors can be easily connected with a small force by cam action between a cam groove and a cam follower according to the rotation of the lever from an initial position to a proper connection position after housings are opposed to each other and initially fit.

[0003] The connector as described above may be provided with a detecting member for detecting a state where the lever has not reached the connection position and the housing has not been properly connected to the mating housing yet (incompletely connected state). The detecting member stays at a standby position in the incompletely connected state of the lever, and is displaceable to a detection position when the lever reaches the connection position. That is, whether or not the lever has reached the connected state can be judged by displacing the detecting member.

PRIOR ART DOCUMENT

Patent Document

[0004] Patent Document 1: JP 2003-133001 A

SUMMARY OF THE INVENTION

Problems to be Solved

[0005] For example, in Patent Document 1, if a force is applied to the lever in a direction to release connection with the lever set at a connection position and a detecting member arranged at a connection detection position, a component of force may act on the detecting member in a direction to displace the detecting member from a detection position to a standby position and the detecting member may be displaced to the standby position.

Means to Solve the Problem

[0006] The present disclosure is directed to a lever-type connector with a connector housing connectable to a mating connector, and a lever rotatable from a connection start position to a connection completion position about a rotary shaft provided on a side surface of the connector housing, the lever including an arm portion to be disposed along the side surface of the connector housing and a connection detecting member mounted on the arm portion, the connection detecting member being allowed to be displaced from a standby position to a detection position when the lever reaches the connection completion position and being restricted from being displaced from the standby position to the detection position when the lever is at an incomplete connection position without reaching the connection completion position, and the arm portion and the connection detecting member being provided with movement restricting portions for restricting the connection detecting member in an inclined posture from being displaced toward the standby position by locking each other if the connection detecting member is set in the inclined posture inclined from a proper posture with respect to the arm portion by applying a force to the arm portion in a direction to release connection with the connection detecting member disposed at the detection position and for allowing a displacement of the connection detecting member if the connection detecting member is set in the proper posture.

Effect of the Invention

[0007] According to the lever-type connector of the present disclosure, even if a force is applied to the lever in a direction to release connection with the connection detecting member arranged at the

detection position, the connection detecting member can be displaced from the detection position to the standby position and the connection release of the connector can be suppressed.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0008] FIG. 1 is a side view of a lever-type connector of a first embodiment (a lever is at a connection start position).

[0009] FIG. 2 is a side view of the lever-type connector of the first embodiment (the lever is at a connection completion position and a connection detecting member is at a standby position).

[0010] FIG. 3 is a section along I-I of FIG. 2.

[0011] FIG. 4 is a side view of the lever-type connector of the first embodiment (the lever is at the connection completion position and the connection detecting member is at a detection position).

[0012] FIG. 5 is a section along II-II of FIG. 4.

[0013] FIG. 6 is a partial enlarged side view of FIG. 4.

[0014] FIG. 7 is a section along III-III of FIG. 6.

[0015] FIG. 8 is a partial enlarged side view when the connection detecting member of the first embodiment is set in an inclined posture.

[0016] FIG. 9 is a section along IV-IV of FIG. 8.

[0017] FIG. 10 is a perspective view of a lever body of the first embodiment.

[0018] FIG. 11 is a perspective view of the connection detecting member of the first embodiment.

[0019] FIG. 12 is a side view of a lever-type connector of a second embodiment (a lever is at a connection start position).

[0020] FIG. 13 is a side view of the lever-type connector of the second embodiment (the lever is at a connection completion position and a connection detecting member is at a standby position).

[0021] FIG. 14 is a section along V-V of FIG. 13.

[0022] FIG. 15 is a side view of the lever-type connector of the second embodiment (the lever is at the connection completion position and the connection detecting member is at a detection position).

[0023] FIG. 16 is a section along VI-VI of FIG. 15.

[0024] FIG. 17 is a partial enlarged side view of FIG. 15.

[0025] FIG. 18 is a section along VII-VII of FIG. 17.

[0026] FIG. 19 is a partial enlarged side view when the connection detecting member of the second embodiment is set in an inclined posture.

[0027] FIG. 20 is a section along VIII-VIII of FIG. 19.

[0028] FIG. 21 is a perspective view of a lever body of the second embodiment.

[0029] FIG. 22 is a perspective view of the connection detecting member of the second embodiment.

DETAILED DESCRIPTION TO EXECUTE THE INVENTION

Summary of Embodiments

[0030] First, embodiments of the present disclosure are listed and described.

[0031] (1) The lever-type connector of the present disclosure is provided with a connector housing connectable to a mating connector, and a lever rotatable from a connection start position to a connection completion position about a rotary shaft provided on a side surface of the connector housing, the lever including an arm portion to be disposed along the side surface of the connector housing and a connection detecting member mounted on the arm portion, the connection detecting member being allowed to be displaced from a standby position to a detection position when the lever reaches the connection completion position and being restricted from being displaced from the standby position to the detection position when the lever is at an incomplete connection position without reaching the connection completion position, and the arm portion and the

connection detecting member being provided with movement restricting portions for restricting the connection detecting member in an inclined posture from being displaced toward the standby position by locking each other if the connection detecting member is set in the inclined posture inclined from a proper posture with respect to the arm portion by applying a force to the arm portion in a direction to release connection with the connection detecting member disposed at the detection position and for allowing a displacement of the connection detecting member if the connection detecting member is set in the proper posture.

[0032] According to the above configuration, a displacement of the connection detecting member at the detection position to the standby position is restricted by the movement restricting portions even if a force is applied to the lever in a connected state in a direction to unintentionally release the connection. Therefore, the connection release of the connector can be suppressed.

[0033] (2) The arm portion may be provided with a window portion penetrating through the arm portion, the connection detecting member may include a body portion disposed along the connector housing and a projecting portion disposed in the window portion by projecting to rise from a plate surface of the body portion, the projecting portion being displaceable between the standby position and the detection position in the window portion, a locking portion projecting in a direction intersecting a displacement direction of the projecting portion may be provided on a connection completion side wall surface disposed at the connection completion position side, out of an inner peripheral surface of the window portion, and the connection detecting member in the inclined posture may be restricted from being displaced toward the standby position by a part of the projecting portion being locked to the locking portion.

[0034] By this configuration, it is possible to provide a specific mode of restricting a displacement of the connection detecting member in the inclined posture to the standby position if a force is applied to the lever in the connected state in a direction to release the connection.

[0035] (3) The locking portion may be ridge-like and provided in a region corresponding to the projecting portion disposed at the detection position, out of the connection completion side wall surface of the window portion, the projecting portion may be provided with a concave locking recess capable of receiving the locking portion when the connection detecting member is set in the inclined posture, and the connection detecting member in the inclined posture may be restricted from being displaced toward the standby position by the locking portion being locked into the locking recess.

[0036] (4) A cut portion capable of receiving a part of the projecting portion in the inclined posture by being recessed from the connection completion side wall surface may be provided in a region corresponding to the projecting portion disposed at the detection position, out of the connection completion side wall surface of the window portion, and the connection detecting member in the inclined posture may be restricted from being displaced toward the standby position by a part of the projecting portion being locked to an opening edge part of the cut portion.

[0037] (5) An arcuate portion projecting while having an arcuate shape centered on the rotary shaft and capable of holding the connection detecting member at the standby position in a rotation process of the lever and a concave escape recess connected to the arcuate portion and allowing the connection detecting member of the lever disposed at the connection completion position to displaceably escape to the detection position may be formed on the side surface of the connector housing.

[0038] By this configuration, it is possible to provide a specific mode of holding the connection detecting member at the standby position in the rotation process and enabling a displacement to the detection position when the connection is completed.

Details of Embodiments of Present Disclosure

[0039] Specific examples of a lever-type connector of the present disclosure are described below with reference to the drawings. Note that the present disclosure is not limited to these illustrations, but is represented by claims and intended to include all changes in the scope of claims and in the

meaning and scope of equivalents.

First Embodiment

[0040] A first embodiment is described with reference to FIGS. **1** to **11**. In the following description, a connection direction (X direction) to a mating connector is a forward direction and an opposite direction thereof is a rearward direction. Further, a Y direction is a rightward direction, and an opposite direction thereof is a leftward direction. Further, concerning a lever, a connection advance direction (rightward direction of FIG. **1**) in a rotation direction is a connection completion direction, and an opposite direction thereof (leftward direction of FIG. **1**) is a connection start direction. Further, for convenience, female terminals and wires are not shown.

[0041] A lever-type connector of this embodiment is a female connector **10** to be connected to a male connector (an example of a mating connector). The female connector **10** is provided with a housing body (an example of a connector housing) **11** for holding the female terminals connected to ends of the wires, a wire cover (an example of the connector housing) **20** to be assembled with the housing body **11** and a lever **30** to be likewise assembled with the housing body **11**.

[Housing Body **11**]

[0042] The housing body **11** is made of synthetic resin and configured such that an inner housing for holding the female terminals is held inside an outer housing **12**. The inner housing is substantially in the form of a rectangular block and includes a plurality of cavities penetrating in the front-rear direction, and the female terminal and an end part of the wire connected to the female terminal can be accommodated in each cavity.

[0043] The outer housing **12** is substantially in the form of a rectangular tube as a whole. As shown in FIGS. **1** and **2**, the outer housing **12** is provided with an inner holding wall **13**, two cover holding walls **14**, two lever holding walls **15** and two rotary shafts **16**. The inner holding wall **13** is substantially in the form of a rectangular tube and can hold the aforementioned inner housing inside.

[0044] The two cover holding walls **14** extend in the front-rear direction while having a U-shaped cross-section when viewed from behind, and are arranged to face each other. As shown in FIGS. **1**, **2** and the like, the cover holding walls **14** are overlapped on the outer periphery of the inner holding wall **13** to surround wall portions on short sides of the inner holding wall **13** facing each other, and extend further rearward (upward in FIG. **1**) from the rear end of the inner holding wall **13**. Locked portions for locking the wire cover **20** are provided on these extending parts.

[0045] The two lever holding walls **15** are arranged apart from the outer periphery of the inner holding wall **13** via spaces on front end sides (lower sides of FIG. **1**) of wall portions on long sides of the inner holding wall **13** facing each other. Both end parts in the front-rear direction of each lever holding wall **15** are coupled to the cover holding walls **14**. In this way, a space open in the front-rear direction is formed between the lever holding wall **15** and the inner holding wall **13**. This space serves as a lever accommodating portion for allowing the entrance of a part of an arm portion **32** of the lever **30** to be described later.

[0046] The two rotary shafts **16** are provided to project toward the inner holding wall **13** from the inner surfaces of the lever holding walls **15** in the lever accommodating portions. The rotary shaft **16** has a cylindrical shape for rotatably supporting the lever **30**.

[Wire Cover **20**]

[0047] The wire cover **20** is a member to be assembled with the housing body **11** to cover a back surface for protecting the wires drawn out from the inner housing and leading out the wires in a predetermined direction. The wire cover **20** is made of synthetic resin and has a substantially box shape with open front and right sides. It is assumed below that a wall extending in a lateral direction to cover the back surface of the housing body **11** is a main wall **21** (see FIG. **3**) and two walls extending forward from side edge parts of the main wall **21** and disposed to face each other are side walls **22**. One end side (left side) of the main wall **21** is curved forward and connected to a left edge part of the side wall **22**.

[0048] An extended wall portion **23** extending forward is provided on the front end of a central part of the side wall **22** in a longitudinal direction (lateral direction, Y direction). As shown in FIGS. **1** and **2**, the extended wall portion **23** has a tapered trapezoidal shape. With the wire cover **20** assembled with the housing body **11**, the extended wall portion **23** is arranged on the outer periphery of the inner holding wall **13**.

[0049] The main wall **21** of the wire cover **20** is provided with a cantilevered resilient locking piece **24** projecting rearward. The resilient locking piece **24** is formed by cutting and raising the main wall **21** into a U shape from left to right, and is resiliently deformable in a direction intersecting a plate surface of the main wall **21**. A right end part of the resilient locking piece **24** comes into contact with a rotating portion **41** of the lever **30** from left at a connection completion position (see FIG. **2**), whereby the lever **30** is restricted from rotating from the connection completion position to a connection start position.

[0050] The wire cover **20** is provided with locking projections provided to project outward on the right front end and of the main wall **21** and the left front ends of the side walls **22**. By fitting these locking projections into the respective locked portions provided in the cover holding walls **14**, the wire cover **20** is fixed to the housing body **11**.

[Lever **30**]

[0051] The lever **30** is a member for assisting the connection and separation of the female connector **10** and a male connector **70**. The lever **30** is made of synthetic resin and, as shown in FIGS. **1**, **2** and the like, provided with a lever body **31** to be rotatably assembled with the housing body **11** (outer housing **12**) and a connection detecting member **51** to be assembled with the lever body **31** for conforming the connection completion of the housing body **11** to the male connector **70**.

[0052] As shown in FIG. **10**, the lever body **31** includes two arm portions **32** and the rotating portion **41** coupling one end sides of the two arm portions **32**. The two arm portions **32** are each in the form of an elongated plate, arranged on both outer sides of the inner holding wall **13** and facing each other (see FIGS. **1** and **3**). A tip part (end part opposite to the rotating portion **41**) of each of the two arm portion **32** is accommodated into each of the two lever accommodating portions and arranged along each of the two lever holding walls **15**. As shown in FIG. **10**, each arm portion **32** includes a shaft hole **33** for receiving the rotary shaft **16**, a gear portion **35** disposed around the shaft hole **33**, a window portion **36** for accommodating the connection detecting member **51** and a slide guiding portion **38** for sliding the connection detecting member **51**.

[0053] The shaft hole **33** is a circular hole disposed in the tip part of the arm portion **32**. A passage recess **34** communicating with the shaft hole **33** and allowing the passage of a retaining projection **17** (see FIG. **1**) provided on the tip of the rotary shaft **16** is disposed on a peripheral edge part of the shaft hole **33**. The gear portion **35** is disposed on the inner surface of the arm portion **32** facing the inner holding wall **13** and includes a first gear tooth **35A** and a second gear tooth **35B** extending to surround the shaft hole **33**. The first and second gear teeth **35A**, **35B** are arranged at an interval.

[0054] The window portion **36** is a through hole arranged at a position closer to the rotating portion **41** than the shaft hole **33** in the arm portion **32**. The window portion **36** has a rectangular shape and is arranged in such an orientation that a longitudinal direction coincides with an extension direction of the arm portion **32**.

[0055] The slide guiding portion **38** rises into a U shape along a hole edge part of the window portion **36** except on the side of the rotating portion **41** on the inner surface (surface facing the opposite arm portion **32**) of the arm portion **32**, and constitutes a groove-like part extending in the extension direction (substantially vertical direction of FIG. **10**) of the arm portion **32**. The inner wall surfaces of groove walls (wall portions disposed to face each other) constituted by the slide guiding portions **38** are continuous and flush with the inner peripheral surfaces of the window portions **36**. The inner wall surface of the slide guiding portion **38** is described to be included in the inner peripheral surface of the window portion **36** below.

[0056] Out of the inner peripheral surface of the window portion **36**, a wall surface (connection completion side wall surface **36A**) arranged on the connection completion position side (right side of FIG. **2**) is provided with a rectangular ridge portion **37** projecting inward of the window portion **36** (see FIGS. **6** and **10**). In this embodiment, the ridge portion **37** is provided at a position continuous and flush with the inner surface (surface on the side of the slide guiding portion **38**) of the arm portion **32** substantially in a central part in the longitudinal direction of the window portion **36** (see FIG. **10**). That is, the ridge portion **37** projects from the inner wall surface of the slide guiding portion **38**.

[0057] As shown in FIG. **10**, the rotating portion **41** includes a first bridge portion **42** and a second bridge portion **43** linking end parts of the two arm portions **32** and a third bridge portion **44** linking the first and second bridge portions **43**, **44** at a middle position between the two arm portions **32**. A space defined by the three bridge portions **42**, **43** and **44** serves as a detecting member accommodating portion **45** for accommodating a later-described push-in portion **53** of the connection detecting member **51**.

[0058] As shown in FIG. **11**, the connection detecting member **51** includes two detecting member side arm portions **52** (an example of a body portion) arranged along side surfaces of the wire cover **20** while facing each other, the push-in portion **53** coupling one end parts of the two detecting member side arm portions **52**, and guide portions **54** and projecting portions **61** respectively projecting from the two detecting member side arm portions **52**. A width of the detecting member side arm portion **52** is substantially equal to that of the aforementioned window portion **36** as a whole.

[0059] The guide portion **54** is an L-shaped projection located near a tip side opposite to the push-in portion **53** on the detecting member side arm portion **52** and projecting from an inner surface facing the opposite detecting member side arm portion **52**. In particular, the guide portion **54** is composed of a guide portion body **55** extending along an extension direction of the detecting member side arm portion **52** on an edge part on a connection completion direction side (right side of FIG. **2**) of the detecting member side arm portion **52** and a contact portion **56** extending from a tip side (side opposite to the push-in portion **53**) of the guide portion body **55** toward an edge part on a connection start direction side (left side of FIG. **2**).

[0060] The guide portion **54** is so set that the inner surface thereof is located further inward than the inner surface of the slide guiding portion **38** with the connection detecting member **51** assembled with the lever body **31** (see FIGS. **3** and **5**).

[0061] The projecting portion **61** is a projection located adjacent to the tip of the detecting member side arm portion **52** and projecting to rise from a plate surface from an outer surface opposite to the inner surface on the detecting member side arm portion **52**. In particular, the projecting portion **61** includes a rising portion **62** rising perpendicularly outward from the plate surface of the detecting member side arm portion **52**, a parallel portion **63** extending in parallel to the plate surface of the detecting member side arm portion **52** from the tip of the rising portion **62** toward the tip side of the detecting member side arm portion **52** and an inclined portion **64** inclined in a direction approaching the plate surface of the detecting member side arm portion **52** from the parallel portion **63** to the tip side of the detecting member side arm portion **52**, and has a trapezoidal side view shape.

[0062] The projecting portion **61** includes a hollow portion surrounded by these rising portion **62**, parallel portion **62** and inclined portion **64** and the guide portion body **55**. This hollow portion serves as a locking recess **67** of a size to be able to receive the aforementioned ridge portion **37** by being concavely recessed from a surface of the detecting member side arm portion **52** on the connection completion side.

[0063] Out of a pair of side edge parts of the projecting portion **61**, the side edge part on a right side (connection completion position side) of FIG. **2** is provided with a cut portion **65** cut along the side edge part from the tip of the detecting member side arm portion **52** as shown in FIGS. **6** and

11. That is, a width of the tip part of the detecting member side arm portion **52** is made somewhat smaller than that of the window portion **36** by this cut portion **65**, and a clearance is present between the tip part of the detecting member side arm portion **52** and the inner peripheral wall of the window portion **36**. This cut portion **65** extends from the tip of the detecting member side arm portion **52** to a position adjacent to the parallel portion **63** of the projecting portion **61**.

[0064] Further, the outer surface of the detecting member side arm portion **52** is continuously cut along the side edge part from an end part of the cut portion **65** on the side of the push-in portion **53** to form a rail portion **66**. That is, the rail portion **66** is a thin plate-like part extending along the side edge part on the inner surface side of the detecting member side arm portion **52**. By moving the aforementioned ridge portion **37** along this rail portion **66**, a smooth displacement of the connection detecting member **51** can be guided.

[0065] As shown in FIGS. **3** and **5**, the lever body **31** and the connection detecting member **51** are assembled such that the push-in portion **53** and parts of the detecting member side arm portions **52** of the connection detecting member **51** can be accommodated in the detecting member accommodating portion **45** of the lever body **31** and the two detecting member side arm portions **52** of the connection detecting member **51** are respectively slidably accommodated in the two slide guiding portions **38** of the lever body **31**. In this state, the guide portions **54** of the connection detecting member **51** are set to project from the inner surfaces of the arm portions **32** (slide guiding portions **38**) of the lever body **31** (see FIGS. **3** and **5**). Further, the two projecting portions **61** respectively project into the two window portions **36**. A projecting direction of the projecting portion **61** coincides with a penetrating direction of the window portion **36**.

[0066] The connection detecting member **51** is displaceable between a standby position (position shown in FIGS. **2** and **3**) where a part of the push-in portion **53** projects from the rotating portion **41** and a detection position (position shown in FIGS. **4** and **5**) where the push-in portion **53** is entirely accommodated in the detecting member accommodating portion **45** and does not project from the rotating portion **41**.

[0067] Note that, with the connection detecting member **51** set at the standby position, the ridge portions **37** projecting from the inner peripheral walls of the window portions **36** are set to overlap the cut portions **65** of the detecting member side arm portions **52** of the connection detecting member **51**.

[0068] Further, with the connection detecting member **51** disposed at the detection position, the ridge portions **37** are set to overlap the rail portions **66** of the connection detecting member **51** from outside, whereby outward movements of the projecting portions **61** are restricted (see FIG. **6**). Further, at this time, the locking recesses **67** provided in the projecting portions **61** are set to be arranged at positions facing the ridge portions **37**.

[0069] The two arm portions **32** of the aforementioned lever **30** are respectively accommodated in the two lever accommodating portions of the housing body **11** as shown in FIGS. **1** and **2**. Tip sides of the outer surfaces of the arm portions **32** are disposed along the lever holding walls **15**. Further, the inner surfaces of the arm portions **32** are disposed along the inner holding wall **13** and the wire cover **20**.

[0070] The lever **30** is rotatable about the rotary shafts **16** between the connection start position (position shown in FIG. **1**) and the connection completion position (position shown in FIG. **2**). When being at the connection start position, the lever **30** is arranged to be inclined obliquely leftward with respect to the housing body **11**. When being at the connection completion position, the lever **30** is arranged to be inclined obliquely rightward with respect to the housing body **11**.

[0071] An arcuate portion **25** having an arcuate shape centered on the rotary shaft **16**, projecting outward and capable of holding the connection detecting member **51** at the standby position in a rotation process of the lever **30** is provided on the side surface of the wire cover **20**. The connection detecting member **51** is so set that the lower ends of the contact portions **56** of the guide portions **54** slide on the arcuate portions **25** during the rotation of the lever **30**, whereby the connection

detecting member **51** is held at the standby position.

[0072] Further, a concave escape recess **26** connected to the arcuate portion **25** for allowing the connection detecting member **51** of the lever **30** disposed at the connection completion position to displaceably escape from the standby position to the detection position is formed to be recessed forward (downward of FIG. **1**) in the side surface of the wire cover **20**. In this way, the connection detecting member **51** of the lever **30** at the connection completion position is displaceable forward (toward the detection position), whereby the connection completion can be detected. Further, by substantially tightly fitting the connection detecting member **51** into the escape recesses **26**, a rotating operation of the lever **30** is hindered.

[Male Connector **70**]

[0073] As shown in FIG. **1**, the male connector **70** (an example of the mating connector) includes a bottom plate portion **71** and a tubular receptacle **72** extending from the bottom plate portion **71**. Male terminals to be connected to the female terminals held in the inner housing are held in the male connector **70**.

[0074] Note that, although not shown, gear receiving portions corresponding to the gear portions **35** respectively provided on the two arm portions **32** of the lever **30** are provided on the outer surface of the receptacle **72**. The male connector **70** and the female connector **10** are connected by cam action of the gear portions **35** and the gear receiving portions caused by the rotating operation of the lever **30**.

[Connection of Female Connector **10** and Male Connector **70**]

[0075] Next, an example of a procedure of connecting the female connector **10** and the male connector **70** configured as described above is described below. First, as shown in FIG. **1**, the housing body **11** of the female connector **10** is lightly connected to the male connector **70** and the tip of the receptacle **72** is inserted into a connection space of the housing body **11** with the lever **30** set at the connection start position.

[0076] At this time, the connection detecting member **51** assembled with the lever body **31** is at the standby position where the push-in portion **53** projects upward from the detecting member accommodating portion **45** of the rotating portion **41**. Further, the connection detecting member **51** is held at the standby position by placing the contact portions **56** of the guide portions **54** of the detecting member side arm portions **52** on the upper surfaces of the arcuate portions **25**.

[0077] Subsequently, the lever **30** is rotated from the connection start position (see FIG. **1**) to the connection completion position (see FIG. **2**). During this rotation, since the contact portions **56** of the guide portions **54** slide on the upper surfaces of the arcuate portions **25**, the connection detecting member **51** is held at the standby position. According to the rotation of the lever **30**, the housing body **11** is relatively pulled toward the male connector **70** and connection proceeds by unillustrated cam action.

[0078] As the connection of the female connector **10** and the male connector **70** proceeds and the lever **30** reaches the connection completion position as shown in FIGS. **2** and **3**, the housing body **11** reaches a proper connection position with respect to the male connector **70**. The rotating portion **41** of the lever **30** comes into contact with the resilient locking piece **24** of the wire cover **20** and gradually pushes down the resilient locking piece **24** immediately before the lever **30** reaches the connection completion position, and rides over the resilient locking piece **24** at the same time as the connection position is reached. In this way, the resilient locking piece **24** resiliently returns and is locked to a surface of the rotating portion **41** of the lever **30** on a side opposite to the rotation direction. In this way, the rotation of the lever **30** toward an opposite side in the rotation direction is restricted and the lever **30** is locked at the connection position.

[0079] Subsequently, the push-in portion **53** of the connection detecting member **51** is pushed into the detecting member accommodating portion **45** of the lever body **31** and the connection detecting member **51** at the standby position is displaced to the detection position (see FIGS. **4** and **5**). At this time, the rail portions **66** of the projecting portions **61** arranged in the window portions **36** of the

lever body **31** relatively move along the ridge portions **37**, whereby a smooth displacement is made from the standby position to the detection position.

[0080] A force may be applied in a direction to release the connection to the lever **30** of the female connector **10** in the connected state where the connection detecting member **51** is disposed at the detection position in this way. If such a force is applied, the posture of the connection detecting member **51** with respect to the lever body **31** may be inclined from a proper posture since the lever body **31** more easily largely moves with respect to the connection detecting member **51** (detecting member side arm portions **52**) fit in the escape recesses **26** of the wire cover **20**. Specifically, the posture of the projecting portion **61** with respect to the window portion **36** is inclined as shown in FIG. **8**.

[0081] However, in this embodiment, the ridge portions **37** provided in the window portions **36** are fit into the locking recesses **67** formed in the projecting portions **61** if the posture of the connection detecting member **51** with respect to the lever body **31** is inclined. Accordingly, even if a component of force acts on the connection detecting member **51** in a direction to displace the connection detecting member **51** toward the standby position, a movement of the connection detecting member **51** from the detection position toward the standby position is restricted by locking structures (an example of movement restricting portions) of these ridge portions **37** and the locking recesses **67**, whereby unintended connection release is suppressed.

[0082] As just described, the lever **30** of this embodiment is configured such that the ridge portions **37** and the locking recesses **67** do not interfere if the connection detecting member **51** is in a proper posture with respect to the lever body **31**, and are locked to each other to restrict relative displacements only when the connection detecting member **51** is set in an inclined posture.

[0083] Next, functions and effects of this embodiment configured as described above are described. The female connector **10** of this embodiment is provided with the housing body **11** connectable to the male connector **70** and the lever **30** rotatable from the connection start position to the connection completion position about the rotary shafts **16** provided on the side surfaces of the wire cover **20** assembled with the housing body **11**. The lever **30** includes the arm portions **32** to be disposed along the side surfaces of the housing body **11** and the connection detecting member **51** mounted on the arm portions **32** and allowed to be displaced from the standby position to the detection position when the lever **30** reaches the connection completion position and restricted from being displacing from the standby position to the detection position when the lever **30** is at an incomplete connection position without reaching the connection completion position. The arm portions **32** and the connection detecting member **51** are provided with the ridge portions **37** and the locking recesses **67** for restricting a displacement of the connection detecting member **51** in an inclined posture toward the standby position by being locked to each other if a force is applied to the arm portions **32** and the connection detecting member **51** is set in the inclined posture inclined from the proper posture with respect to the arm portions **32** with the connection detecting member **51** disposed at the detection position.

[0084] According to the above configuration, even if the connection detecting member **51** is set in the inclined posture inclined from the proper posture with respect to the arm portions **32** and a component of force acts on the connection detecting member **51** in a direction to displace the connection detecting member **51** toward the standby position by applying a force to the arm portions **32** in a direction to release the connection with the connection detecting member **51** disposed at the detection position, a displacement of the connection detecting member **51** to the standby position is restricted by the locking structures (movement restricting portions) to the arm portions **32**.

[0085] Further, the arm portions **32** are provided with the window portions **36** penetrating through the arm portions **32**, the connection detecting member **51** includes the detecting member side arm portions **52** to be disposed along the housing body **11** and the projecting portions **61** disposed in the window portions **36** by projecting to rise from the plate surfaces of the detecting member side arm

portions **52**. The projecting portion **61** is displaceable between the standby position and the detection position in the window portion **36**. The ridge portion **37** projecting in the direction intersecting a displacement direction of the projecting portion **61** is provided on the connection completion side wall surface **36A** disposed on the connection completion position side, out of the inner peripheral surface of the window portion **36**, and the connection detecting member **51** in the inclined posture is restricted from being displaced toward the standby position by parts of the projecting portions **61** being locked to the ridge portions **37**.

[0086] Furthermore, the ridge portion **37** is provided in a region corresponding to the projecting portion **61** disposed at the detection position, out of the connection completion side wall surface **36A** of the window portion **36**, the projecting portion **61** is provided with the concave locking recess **67** capable of receiving the ridge portion **37** when the connection detecting member **51** is set in the inclined posture, and the connection detecting member **51** in the inclined posture is restricted from being displaced toward the standby position by the ridge portions **37** being locked into the locking recesses **67**.

[0087] By this configuration, a displacement of the connection detecting member **51** in the inclined posture toward the standby position can be specifically suppressed.

[0088] Further, the side surface of the wire cover **20** is formed with the arcuate portion **25** projecting while having an arcuate shape centered on the rotary shaft **16** and capable of holding the connection detecting member **51** at the standby position in the rotation process of the lever **30** and the concave escape recess **26** connected to the arcuate portions **25** and allowing the connection detecting member **51** of the lever **30** disposed at the connection completion position to displaceably escape to the detection position.

[0089] By this configuration, the connection detecting member **51** of the lever **30** being rotated can be held at the standby position, and the connection detecting member **51** of the lever **30** disposed at the connection completion position can be made movable to the detection position and held at the detection position.

Second Embodiment

[0090] Next, a second embodiment is described using FIGS. **12** to **22**. Note that only a configuration different from the first embodiment is described below and components similar to those of the first embodiment are denoted by the same reference signs and repeated description is omitted.

[0091] A female connector **120** (an example of the lever-type connector) of this embodiment differs from that of the first embodiment in locking structures (structures of the movement restricting portions) of a lever body **81** and a connection detecting member **101** when the connection detecting member **101** is set in an inclined posture with respect to a lever body **81**. Specifically, the locking structure of a projecting portion **111** to a window portion **86** is different from that of the first embodiment.

[0092] The window portion **86** of the lever body **81** of this embodiment is formed by concavely cutting a tip side (end side opposite to a rotating portion **91**) of a connection completion side wall surface **86A**, out of the inner peripheral surface of the rectangular window portion **86** while obliquely expanding the window portion **86** in diameter toward the tip side as shown in FIG. **21**. An opening of this cut portion **96** is set at a position, where a rising portion **112** of the projecting portion **111** disposed at a detection position can enter the cut portion **96**, on the connection completion side wall surface **86A** of the window portion **86**. That is, the cut portion **96** is provided in a region corresponding to the projecting portion **111** disposed at the detection position. In the following description, a part of the window portion **86** located on the side of the rotating portion **91** is referred to as a narrow portion **97**, and a part thereof located on the tip side and formed to be relatively wider than the narrow portion **97** by the cut portion **96** is referred to as a wide portion **98**.

[0093] A width of a detecting member side arm portion **102** of the connection detecting member **101** of this embodiment is substantially equal to that of the narrow portion **97** of the window

portion **86** as a whole. That is, the width of the detecting member side arm portion **102** is smaller than that of the wide portion **98**. Further, as in the first embodiment, an L-shaped and projecting guide portion **104** is provided on the inner surface of the detecting member side arm portion **102**. [0094] In the female connector **120** of this embodiment, a force may be applied to a lever **80** in a direction to release connection with the connection detecting member **101** disposed at the detection position and the posture of the connection detecting member **101** with respect to the lever body **81** may be set to an inclined posture inclined from a proper posture. Specifically, the posture of the projecting portion **111** with respect to the window portion **86** may be inclined as shown in FIG. **19**. [0095] However, in this embodiment, if the posture of the connection detecting member **101** with respect to the lever body **81** is inclined, a part of the projecting portion **111** is fit into the cut portion **96** while being inclined, and the rising portion **112** of the projecting portion **111** is locked to a boundary part between the narrow portion **97** and the wide portion **98**, i.e. an opening edge part **96A** of the cut portion **96**. Therefore, even if a force acts on the connection detecting member **101** in a direction to displace the connection detecting member **101** toward a standby position, the connection detecting member **101** is restricted from moving from the detection position toward the standby position by locking structures (an example of the movement restricting portions) of the projecting portions **111** and the opening edge parts (boundary parts between the narrow portions **97** and the wide portions **98**) **96A** of the cut portions **96**, whereby unintended connection release is suppressed.

[0096] As just described, according to the female connector **120** of this embodiment, the cut portion **96** recessed from the connection completion side wall surface **86A** and capable of receiving a part of the projecting portion **111** in the inclined posture is provided in a region corresponding to the projecting portion **111** disposed at the detection position, out of the connection completion side wall surface **86A** of the window portion **86**, whereby a region (wall surface of the narrow portion **97**) except the cut portion **96** projects into the window portion **86**. By locking the connection detecting member **101** in the inclined posture to the opening edge parts (boundary parts between the narrow portions **97** and the wide portions **98**) **96A** of the cut portions **96**, a displacement toward the standby position can be restricted.

[0097] Further, as in the first embodiment, the lever **80** of this embodiment is configured such that the projecting portions **111** and the opening edge parts **96A** of the cut portions **96** do not interfere if the connection detecting member **101** is set in the proper posture with respect to the lever body **81**, and are locked to each other to restrict relative displacements only when the connection detecting member **101** is set in an inclined posture.

[0098] Further, according to this embodiment, since only the lever body **81** needs to be changed to include the cut portions **96** in the window portions **86** while the conventional connection detecting member **101** is used, production cost is excellent.

Other Embodiments

[0099] The present disclosure is not limited to the above described and illustrated embodiments. For example, the following embodiments are also included in the technical scope.

[0100] (1) Although the movement restricting portion is provided on the inner surface on the connection completion position side, out of the inner peripheral surface of the window portion, in the above embodiments, a movement restricting portion having a similar configuration can be provided also on the inner surface on the connection start position side.

[0101] (2) Although the connection detecting member **51** is restricted from being displaced toward the standby position by locking the ridge portions **37** into the locking recesses **67** of the projecting portions **61** in the inclined posture in the first embodiment, the ridge portions can be, for example, directly locked to the rising surfaces of the projecting portions.

[0102] (3) Although the projecting portion **61** of the connection detecting member **51** projects in the rising direction from the plate surface of the detecting member side arm portion **52** in the embodiment, the projecting portion may project in a direction along the plate surface of the

detecting member side arm portion and be locked to a locking portion provided on the arm portion.

LIST OF REFERENCE NUMERALS

[0103] **10, 120**: female connector (lever-type connector) [0104] **11** housing body (connector housing) [0105] **12**: outer housing [0106] **13**: inner holding wall [0107] **14**: cover holding wall [0108] **15**: lever holding wall [0109] **16**: rotary shaft [0110] **17**: retaining projection [0111] **20**: wire cover (connector housing) [0112] **21**: main wall [0113] **22**: side wall [0114] **23**: extended wall portion [0115] **24**: resilient locking piece [0116] **25**: arcuate portion [0117] **26**: escape recess [0118] **30, 80**: lever [0119] **31, 81**: lever body [0120] **32**: arm portion [0121] **33**: shaft hole [0122] **34**: passage recess [0123] **35**: gear portion [0124] **35A**: first gear tooth [0125] **35B**: second gear tooth [0126] **36, 86**: window portion [0127] **36A, 86A**: connection complete side wall surface [0128] **37**: ridge portion (locking portion, movement restricting portion) [0129] **38**: slide guiding portion [0130] **41, 91**: rotating portion [0131] **42**: first bridge portion [0132] **43**: second bridge portion [0133] **44**: third bridge portion [0134] **45**: detecting member accommodating portion [0135] **51, 101**: connection detecting member [0136] **52, 102**: detecting member side arm portion (body portion) [0137] **53**: push-in portion [0138] **54, 104**: guide portion [0139] **55**: guide body portion [0140] **56**: contact portion [0141] **61, 111**: projecting portion [0142] **62, 112**: rising portion [0143] **63** parallel portion [0144] **64**: inclined portion [0145] **65**: cut portion [0146] **66**: rail portion [0147] **67** locking recess (movement restricting portion) [0148] **70**: male connector (mating connector) [0149] **71** bottom plate portion [0150] **72**: receptacle [0151] **96** cut portion [0152] **96A**: opening edge part [0153] **97** narrow portion [0154] **98**: wide portion

Claims

1. A lever-type connector, comprising: a connector housing connectable to a mating connector; and a lever rotatable from a connection start position to a connection completion position about a rotary shaft provided on a side surface of the connector housing, the lever including an arm portion disposed along the side surface of the connector housing and a connection detecting member mounted on the arm portion, the connection detecting member being allowed to be displaced from a standby position to a detection position when the lever reaches the connection completion position and being restricted from being displaced from the standby position to the detection position when the lever is at an incomplete connection position without reaching the connection completion position, and the arm portion and the connection detecting member being provided with movement restricting portions for restricting the connection detecting member in an inclined posture from being displaced toward the standby position by locking each other if the connection detecting member is set in the inclined posture inclined from a proper posture with respect to the arm portion by applying a force to the arm portion in a direction to release connection with the connection detecting member disposed at the detection position and for allowing a displacement of the connection detecting member if the connection detecting member is set in the proper posture.
2. The lever-type connector according to claim 1, wherein: the arm portion is provided with a window portion penetrating through the arm portion, the connection detecting member includes a body portion to be disposed along the connector housing and a projecting portion disposed in the window portion by projecting to rise from a plate surface of the body portion, the projecting portion being displaceable between the standby position and the detection position in the window portion, a locking portion projecting in a direction intersecting a displacement direction of the projecting portion is provided on a connection completion side wall surface disposed at the connection completion position side, out of an inner peripheral surface of the window portion, and the connection detecting member in the inclined posture is restricted from being displaced toward the standby position by a part of the projecting portion being locked to the locking portion.
3. The lever-type connector according to claim 2, wherein: the locking portion is ridge-like and provided in a region corresponding to the projecting portion disposed at the detection position, out

of the connection completion side wall surface of the window portion, the projecting portion is provided with a concave locking recess capable of receiving the locking portion when the connection detecting member is set in the inclined posture, and the connection detecting member in the inclined posture is restricted from being displaced toward the standby position by the locking portion being locked into the locking recess.

4. The lever-type connector according to claim 2, wherein: a cut portion capable of receiving a part of the projecting portion in the inclined posture by being recessed from the connection completion side wall surface is provided in a region corresponding to the projecting portion disposed at the detection position, out of the connection completion side wall surface of the window portion, and the connection detecting member in the inclined posture is restricted from being displaced toward the standby position by a part of the projecting portion being locked to an opening edge part of the cut portion.

5. The lever-type connector according to claim 1, wherein an arcuate portion projecting while having an arcuate shape centered on the rotary shaft and capable of holding the connection detecting member at the standby position in a rotation process of the lever and a concave escape recess connected to the arcuate portion and allowing the connection detecting member of the lever disposed at the connection completion position to displaceably escape to the detection position are formed on the side surface of the connector housing.
